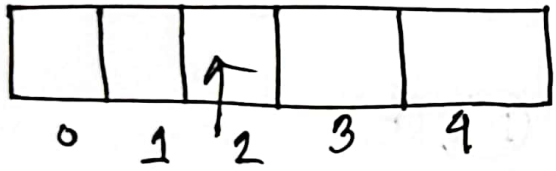
TREE

09-08-23

Definition

1) non-linear
2) hierarchical
3) acyclic → must !
4) Connected
5) There is only (a) path from root to a node
6) A node has (a) single parent

Data structure

a = 

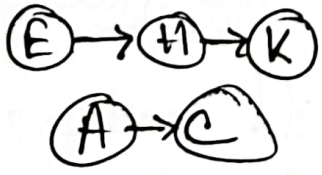
→ linearity

(#) 10 → 15 → 7 → 34 → None

(#) tree is based on Non-linearity → can add elements anywhere.

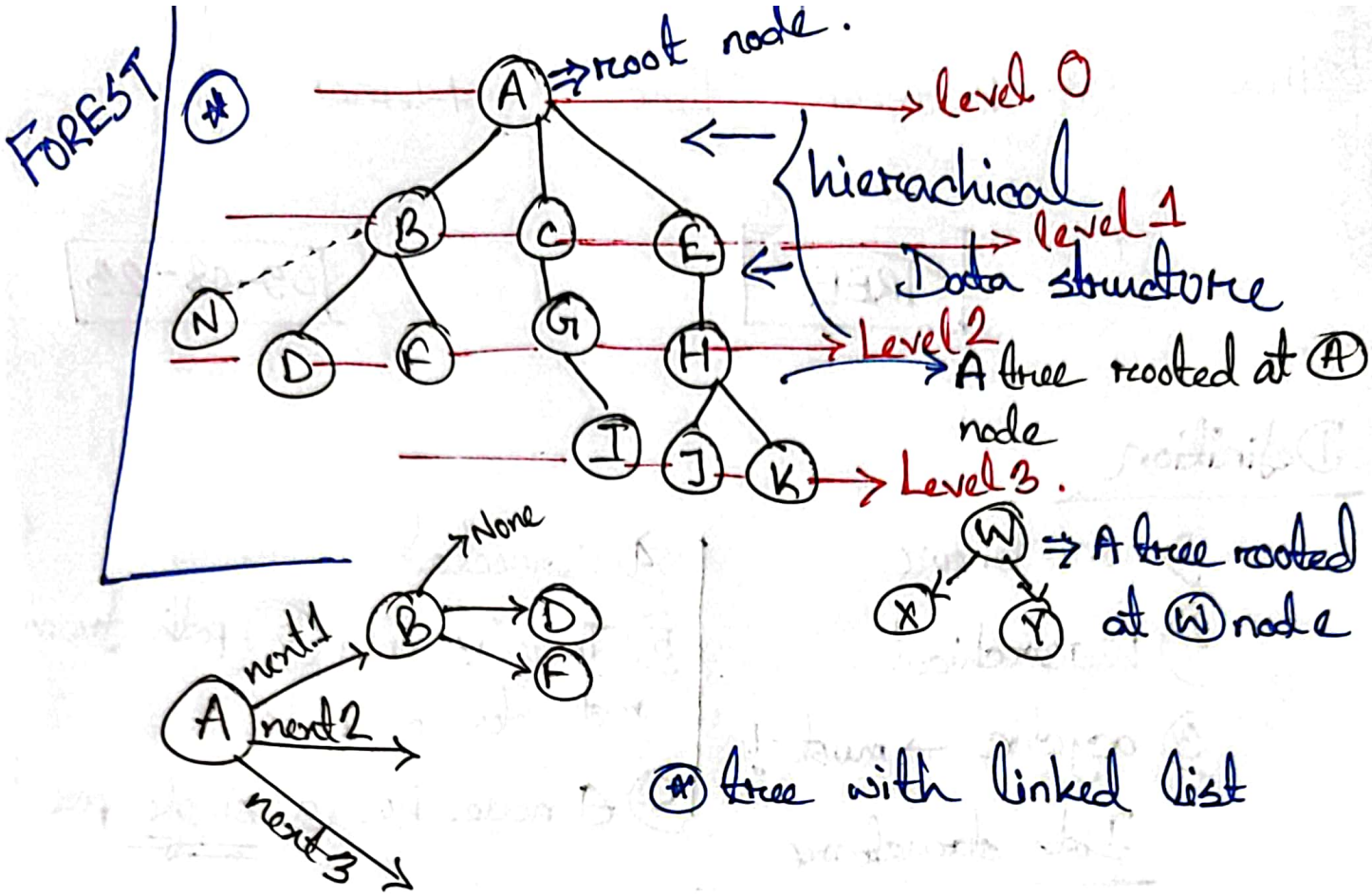
(#) path edge sequence from a node to another node

Ex:



↳ next page

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1) Node/Vertex: A, B, D, F

2) Edge: the line directly connecting two nodes

Adjacent nodes are connected by edge.

Ex: A, C → adjacent nodes

A, H → non-adjacent nodes

3) Parent Node: E is the parent node of H

4) Child Node: G is the child node of C

B, C, E - - - - A

5) Root node:

A node that doesn't have a parent node.

6) Leaf Node:

Nodes that don't have any child nodes.

7) Internal Node:

A node that has children.

Ex: (A), (B), (C), (E), (G), (H) → Internal nodes.

8) Sibling Node:

Nodes that have same parent.

Ex: (B), (C), (E) → same parent exist

(J), (K) → " "

9) Grandparent Node:

C is the grandparent of I.

10) Uncle Node: B, E are uncle nodes of G.

P.T.O.

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(11) Depth/Level: # of edges.

between the root node & a node.

Ex: Depth of node → (G): 2

" " " → (K): 3

" " " → (A): 0

(12) Height of a node:

max distance from the node to a leaf node.

Ex: Height of node (E):

max (1, 2, 2)

= 2

Height of node (A): ≃ height of tree

max (2, 2, 3, 2, 3, 3, 1)

= 3

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Degree:

of children a node has.

Degree of (A) : 4

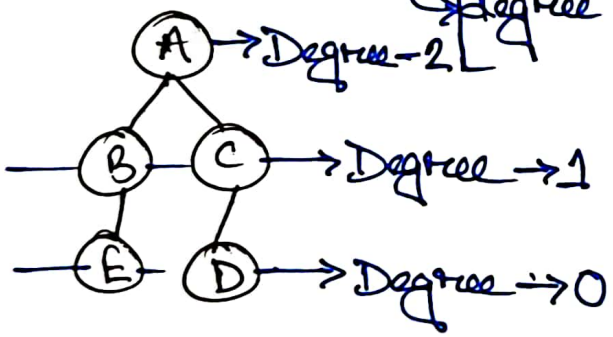
" " (C) : 1

M-ary Tree:

a tree where each node has max (m) children.

m=2 ⇒ Binary Tree

Degree of each node max : 2



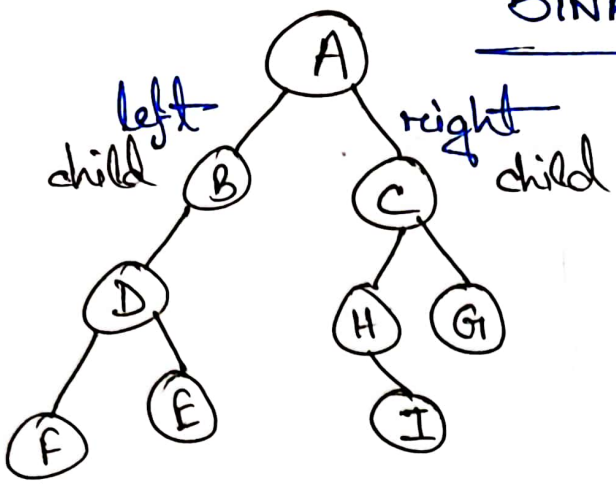
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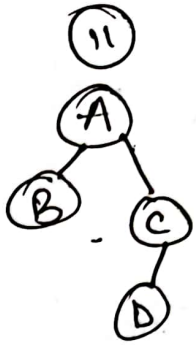
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BINARY TREE

(1)



(11)




```
class Node:
    def __init__(self, elem, right=None, left=None):
        self.elem, self.right, self.left = elem, right,
                                           left
```

#driver code: for (11) →

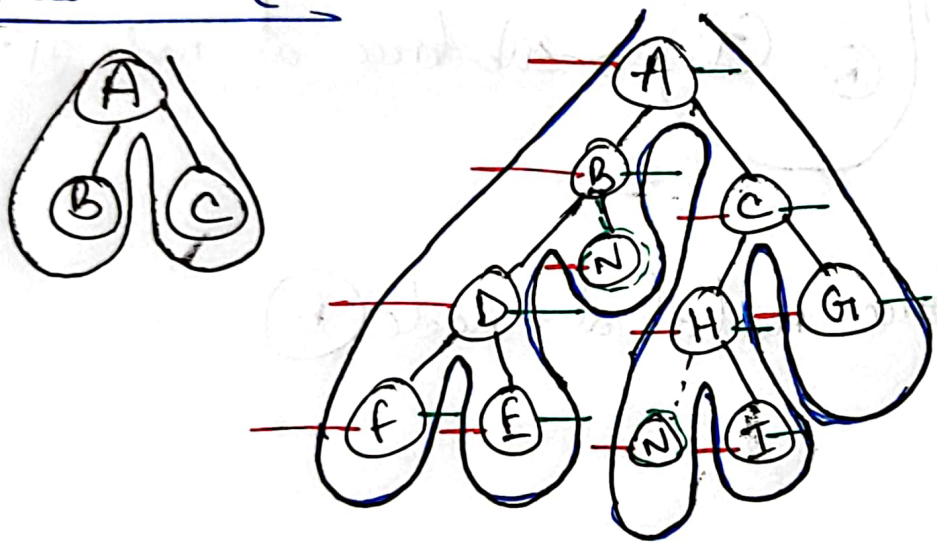
```
node4 = Node("D")
node3 = Node("C", None, "D")
node2 = Node("B")
node1 = Node("A", node3, node2)
root = node1.
```

(A) [Exam Must !!]

Tree Traversal:

1) Pre order ⇒ root left right
2) Post order ⇒ left right root
3) In order ⇒ Left root right

Pre Order → will enclose every child nodes.



pre-order

A B D F E C H I G → outputs (when blue line touches the red dots)

post order

F E D B I H G C A → green dots

In order

F D E B A H I C G.

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Sub-tree

